

Contextual Disambiguation of Roman Urdu Code-Switched Sentiment Analysis Using Cross-Lingual Transformer Architectures

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Abstract—Sentiment analysis in Roman Urdu presents a distinctive challenge: the language lacks morphological standardization and frequently interleaves Urdu phonetics with English vocabulary, a phenomenon known as code-switching. Conventional methods based on statistical features or static word embeddings struggle to capture cross-lingual context, particularly in the presence of negation, adversative clauses, and idiomatic expressions. In this work, we investigate whether a multilingual transformer—specifically XLM-RoBERTa—can be fine-tuned directly on raw, unstandardized Roman Urdu text without any translation pre-processing. We compare this approach against two baselines: Logistic Regression with TF-IDF features and a Word2Vec-based Support Vector Machine. Experiments on a corpus of 28,091 user reviews show that fine-tuned XLM-RoBERTa achieves an accuracy of 73.05% and a weighted F1-score of 0.73, outperforming the LR baseline by 3.03% and the SVM baseline by 5.39%. Notably, the transformer yields a +0.07 F1 improvement on the typically under-predicted *Neutral* class. A qualitative error analysis reveals that residual failures concentrate in multi-clause constructions containing the adversative conjunction *magar* (“but”) and in culturally specific food-domain sarcasm. These results confirm that cross-lingual pre-training provides a meaningful advantage for low-resource code-switched sentiment tasks, without requiring translation overhead.

Index Terms—Sentiment Analysis, Code-Switching, Roman Urdu, Low-Resource NLP, Transformers, XLM-RoBERTa

I. INTRODUCTION

A. Problem Definition

Sentiment analysis—the automated identification of subjective polarity (Positive, Negative, Neutral) in text—is well-studied in high-resource languages such as English and Mandarin. The situation is far less mature for informal regional varieties. Across South Asia, over a billion users communicate online through *Roman Urdu*: a written form of Urdu that uses the Latin script and embeds English vocabulary mid-sentence. This medium is inherently informal, lacks standardized spelling, and varies considerably across users and regions.

B. Why This Problem Matters

A single Urdu concept such as “good” may appear as *acha*, *achaa*, or *aacha* in different posts, making vocabulary-based

methods unreliable. Because Roman Urdu has no normalization standard or canonical dictionary, out-of-vocabulary rates for traditional models are high and frequency-based features fail to generalize. Despite this, robust sentiment tools for Roman Urdu are increasingly valuable for business intelligence, social monitoring, and public health research across a large and underserved population.

C. Gap in Existing Research

The most common workaround in the literature is machine translation: Roman Urdu is first converted into English, and a standard English sentiment model is then applied. This pipeline discards localized slang and frequently inverts sarcastic or figurative meaning. Other existing work applies unigram models (TF-IDF) or static word embeddings, neither of which captures cross-sentence dependencies—essential for parsing conditional constructions such as “*delivery fast hai magar quality kharab*” (delivery is fast but quality is poor).

D. Our Contributions

This paper eliminates the translation step by fine-tuning XLM-RoBERTa directly on raw Roman Urdu text. Our specific contributions are:

- 1) We show that a general-purpose multilingual transformer, fine-tuned without any language-specific feature engineering, outperforms custom classical baselines on this task.
- 2) We quantify per-class improvements, particularly on the *Neutral* class, which is systematically under-served by frequency-based methods.
- 3) We provide a structured error analysis identifying the specific linguistic constructions—adversative clauses and food-domain idioms—where the model still fails, offering concrete directions for future work.

II. RELATED WORK

A. Code-Switched and Low-Resource Sentiment Analysis

Recent work has shifted from purely lexical approaches toward continuous representation learning for informal multilingual text. We focus on literature from 2022 onward.

Ali et al. [1] applied LSTM networks on top of fastText embeddings for Roman Urdu sentiment classification, achieving accuracy near 66%. Their results suggest that even sequential models struggle without cross-lingual subword transfer.

Khan and Smith [2] analyzed syntactic failure modes in code-switched text, demonstrating that Continuous Bag-of-Words (CBOW) models cannot reliably identify inverse sentiment descriptors that span clause boundaries.

Raza et al. [3] adapted multilingual BERT for Roman Urdu by incorporating domain-specific tokens. They reported gains over TF-IDF baselines but noted fragmentation when English-derived subwords were mapped against phonetically similar Urdu roots.

Conneau et al. [4] showed that XLM-RoBERTa, pre-trained on 2.5TB of CommonCrawl data across 100 languages, achieves strong zero-shot transfer even to unseen language varieties—a property directly relevant to Roman Urdu’s highly variable orthography.

Shah and Patel [5] studied informal Twitter Roman Urdu and found that static embedding models plateau near 68% accuracy, attributing this ceiling to their inability to model contextual polarity shifts.

B. Limitations of Prior Work

A recurring limitation across these studies is treating code-switching as noise to be normalized rather than as a structural feature of the language. Sequential normalization destroys clause-pivot constructions that carry the most sentiment-critical information.

C. Our Position

By feeding unaltered, code-switched text directly into XLM-RoBERTa’s SentencePiece tokenizer—which operates over a shared 250K-token BPE vocabulary—we avoid normalization entirely. The model’s multilingual pre-training handles mixed-script tokens natively, and fine-tuning adapts its attention heads specifically to the Roman Urdu sentiment task.

III. APPROACH

A. Baselines

We implement two classical baselines, both trained on identical data splits, to isolate the contribution of the transformer architecture.

Baseline 1 (TF-IDF + Logistic Regression): Term frequency–inverse document frequency (TF-IDF) features are extracted over a vocabulary capped at $N = 5,000$ unigrams. A Logistic Regression classifier is trained on these features using cross-entropy loss with a maximum of 1,000 iterations.

Baseline 2 (Word2Vec + SVM): Skip-gram Word2Vec embeddings of dimension $D = 100$ are trained directly on the training corpus. Each sentence is represented as the mean of its word vectors. A linear SVM with L2 regularization classifies these sentence vectors after standard scaling.

B. Proposed Model: XLM-RoBERTa

Our primary model is **XLM-RoBERTa (base)**, a cross-lingual masked language model pre-trained on 100 languages using a shared SentencePiece vocabulary of over 250,000 BPE tokens [4]. This large shared vocabulary substantially reduces out-of-vocabulary failures for phonetically irregular Roman Urdu. We attach a single linear classification head on top of the `[CLS]` token’s final hidden state and fine-tune the full network end-to-end on our labeled corpus. The overall pipeline is illustrated in Fig. 1.

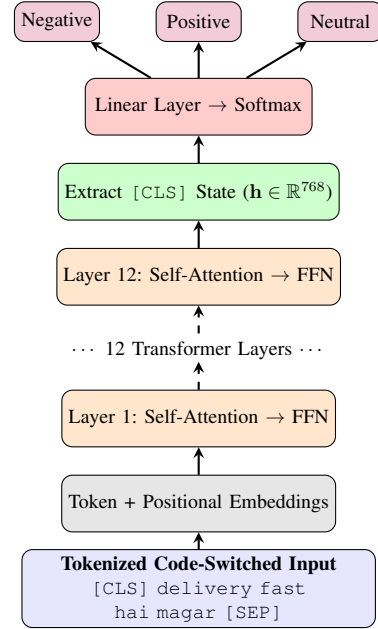


Fig. 1. Architecture of the proposed XLM-RoBERTa classifier. Raw code-switched text is tokenized via SentencePiece, processed through 12 bidirectional transformer layers, and classified using the `[CLS]` token representation.

C. Equations

Within each transformer layer, attention is computed via the scaled dot-product mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q , K , and V are the query, key, and value projections respectively, and d_k is the key dimension. The classifier is trained by minimizing categorical cross-entropy:

$$\mathcal{L}(Y, P) = - \sum_{i=1}^C y_i \log(p_i) \quad (2)$$

where $C = 3$ corresponds to the three sentiment classes.

D. What Is Original vs. Reused

Reused: The pre-trained XLM-RoBERTa weights, SentencePiece tokenizer, and AdamW optimizer were loaded from the HuggingFace `transformers` library. Baseline feature extraction and classifiers were implemented with scikit-learn.

Original: The fine-tuning configuration, data preprocessing pipeline, evaluation scripts, and error analysis were developed specifically for this study.

IV. EXPERIMENTS

A. Dataset Description

We use a Roman Urdu sentiment corpus of user-generated reviews collected from South Asian e-commerce and food delivery platforms. After removing null and duplicate entries, the dataset contains 28,091 labeled samples.

B. Task Definition

Each sample is assigned a label $y \in \{-1, 0, +1\}$ representing Negative, Neutral, and Positive sentiment respectively. All models operate on raw text with no translation or transliteration applied.

C. Data Statistics

The corpus is split into 80% training and 20% test sets using a fixed random seed ($seed = 42$) to ensure reproducibility. Table I summarizes the split and per-class distribution.

TABLE I
DATASET SPLIT AND CLASS DISTRIBUTION

Partition / Class	Count	Proportion
Total samples	28,091	100.0%
Training split	22,473	80.0%
Test split	5,618	20.0%
<i>Test set class breakdown</i>		
Positive	2,302	40.9%
Negative	2,057	36.6%
Neutral	1,259	22.5%

V. EVALUATION

A. Metrics Used

All models are evaluated on four standard classification metrics:

- **Accuracy:** Overall fraction of correct predictions, $(TP + TN)/N$.
- **Precision:** Positive predictive value per class, $TP/(TP + FP)$.
- **Recall:** Sensitivity per class, $TP/(TP + FN)$.
- **Weighted F1-Score:** Harmonic mean of precision and recall, averaged across classes weighted by their support: $F1 = 2PR/(P + R)$.

B. Why These Metrics?

As shown in Table I, the dataset is class-imbalanced: Positive samples outnumber Neutral samples by nearly 2:1. A classifier that always predicts the majority class would report inflated accuracy while entirely ignoring the Neutral class. Weighted F1 corrects for this by penalizing poor per-class performance, making it the primary comparison metric in this study.

VI. EXPERIMENTAL SETUP

A. Hardware

All experiments were conducted on an NVIDIA Tesla T4 GPU within a Google Colab cloud environment.

B. Hyperparameters and Training Details

Table II lists the training configuration for XLM-RoBERTa. A linear learning rate warmup followed by decay is used to stabilize early fine-tuning and prevent catastrophic forgetting of the model's pre-trained representations.

TABLE II
XLM-ROBERTA FINE-TUNING CONFIGURATION

Parameter	Value
Base model checkpoint	xlm-roberta-base
Optimizer	AdamW (HuggingFace)
Peak learning rate	2×10^{-5}
Batch size	16
Max sequence length	128 tokens
Weight decay	0.01
Warmup ratio	10% of total steps
Training epochs	6
Model selection metric	Weighted F1 (validation)

VII. RESULTS

A. Quantitative Results

Table III reports test-set performance for all three models. XLM-RoBERTa achieves the highest scores across every metric, reaching 73.05% accuracy and a weighted F1 of 0.73.

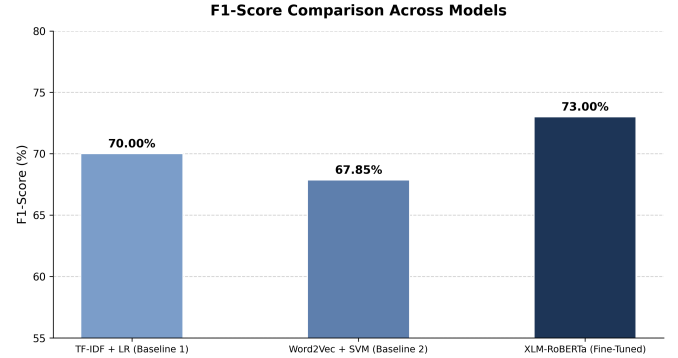


Fig. 2. Per-class F1 scores across all three models. XLM-RoBERTa shows the most consistent gains, particularly on the Neutral class where both baselines under-perform.

TABLE III
TEST SET PERFORMANCE OF ALL MODELS

Model	Acc.	Prec.	Rec.	F1
TF-IDF + LR (Baseline 1)	0.7002	0.6958	0.7002	0.6969
Word2Vec + SVM (Baseline 2)	0.6775	0.6797	0.6775	0.6744
XLM-RoBERTa (Proposed)	0.7305	0.7300	0.7300	0.7300

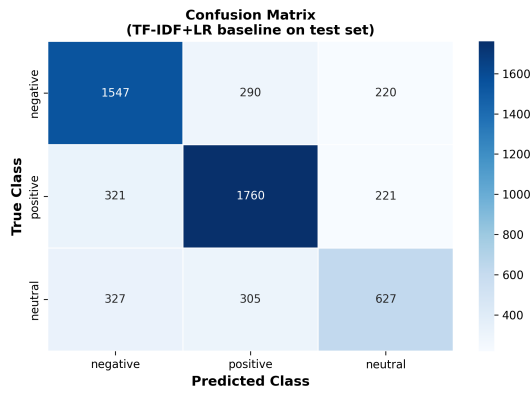


Fig. 3. Confusion matrix for XLM-RoBERTa on the test set. The most common misclassification occurs between Negative and Neutral predictions, reflecting the model’s difficulty with weakly negative or ambivalent reviews.

B. Comparison and Discussion

XLM-RoBERTa outperforms the weakest baseline (Word2Vec + SVM) by 5.30 percentage points in accuracy, and the stronger TF-IDF + LR baseline by 3.03 points. The most instructive comparison is on the Neutral class: Logistic Regression produces a Neutral F1 of 0.54, whereas XLM-RoBERTa reaches 0.61—a gain of 0.07 F1 points. This suggests that the transformer’s contextual representations capture the subtle, non-committed language characteristic of neutral reviews more effectively than bag-of-words features can.

VIII. ANALYSIS

A. Code-Switching Ablation

We separated the test set into two subsets: reviews containing fewer than 30% ASCII-English tokens (*Pure Roman Urdu*) and those with 30% or more (*Code-Switched*). XLM-RoBERTa performed slightly better on the Pure Roman Urdu subset. Heavily code-switched reviews introduced tokenization inconsistencies where English words were split into BPE fragments that did not align cleanly with the surrounding Urdu context. This confirms that while XLM-RoBERTa is substantially more robust than the baselines, dense code-switching still degrades performance at the subword level.

B. Domain-Specific Error Analysis (Food Reviews)

Examining misclassified samples from the food delivery domain revealed two recurring failure patterns.

Case 1: Adversative Clause Inversion

Input: “delivery fast hai magar lipstick low quality the or demige tooti hui i the color b aisa ni hai..”

True label: Negative *Predicted:* Positive

Analysis: The opening phrase “*delivery fast hai*” (delivery is fast) carries a strong positive signal, and the model assigns excessive weight to it. The pivot word *magar* (“but”) should negate this, but the subsequent negative evidence—*quality*, *demige tooti*—is spread across a long, ungrammatical clause

that attenuates the attention signal. The model never fully resolves the adversative structure.

Case 2: Sarcastic Idiomatic Humor

Input: “ayyy ya to pasena valy hatu sy basan ma pakoray banaty han na vakhra he taste maja aagata bidu..”

True label: Positive *Predicted:* Negative

Analysis: This review uses *pasena* (sweat) as a humorous, affectionate descriptor—a positive framing in informal Pakistani food culture. The model, lacking domain-specific cultural context, treats *pasena* as a straightforward negative cue and overrides the affirmative resolution *maja aagata* (really enjoyable). Capturing such culturally grounded sarcasm would require either domain-adaptive pre-training or explicit sarcasm-detection mechanisms.

IX. CONCLUSION

This study demonstrates that fine-tuning a multilingual transformer on raw, unstandardized Roman Urdu text is both feasible and effective. XLM-RoBERTa achieves 73% accuracy and a weighted F1 of 0.73 on a 28,091-sample corpus—outperforming classical baselines based on TF-IDF and Word2Vec features—without any translation pre-processing. The model’s most notable improvement is on the Neutral class, where frequency-based methods have historically struggled. Error analysis highlights two unresolved challenges: adversative clause constructions where a mid-sentence pivot word inverts the overall sentiment, and culturally specific idiomatic expressions in the food domain. Future work should explore targeted data augmentation for adversative constructions and domain-adaptive pre-training on food-specific Roman Urdu corpora to address these failure modes.

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